

A top-down view of a workspace. In the top left, a silver laptop is partially visible. To its right is a white computer mouse. Above the mouse is a cup of dark coffee. In the bottom left, there is a dark brown leather notebook with a white pen resting on it. Next to the notebook are a pair of tortoiseshell glasses. The background is a light teal color with three overlapping teal squares of different shades.

AIClub





Data Acquisition & Basics of Data

What is Data Acquisition?

- Data Acquisition is the process of collecting relevant data, selecting useful data, organizing data properly.
- It comes after Problem Scoping.
- The quality and type of data collected directly affect AI accuracy.
- Without proper data acquisition, AI cannot learn correctly.



What Kind of Data Do We Collect?

During Data Acquisition, we collect raw facts and information needed to solve the problem.

- Data may include numbers (marks, temperature)
- Text (feedback, reviews)
- Images (photos, X-rays)
- Audio (voice recordings).



The type of problem decides the type of data required.

What are Data Formats? (In Data Acquisition)

- Data Formats refer to how collected data is stored so that computers can process it.
- Data must be stored in a proper format.
- Correct format makes data easy to analyze.
- Choosing the right format is part of Data Acquisition.
- Example: Marks may be stored in Excel, Images may be stored as JPG.



Different Formats of Data (Collected During Data Acquisition)

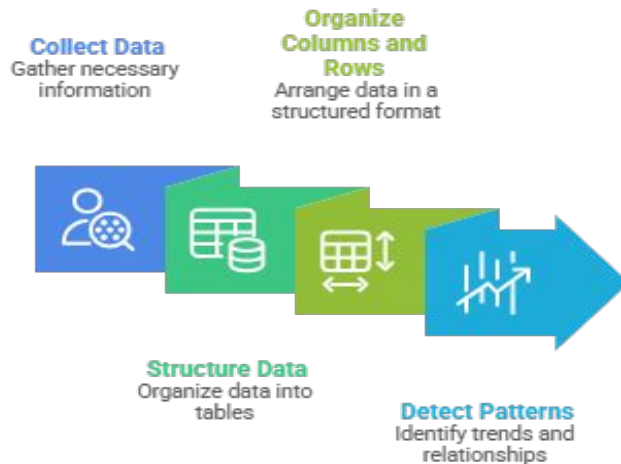
- **Tabular Data (CSV, Excel)** are stored in rows and columns, example: Marks sheet, attendance.
- **Image Data (JPG, PNG)** are used for face recognition, object detection.
- **Audio Data (MP3, WAV)** are used for speech recognition.
- **Video Data (MP4, AVI)** are used for surveillance systems.
- **Text Data (TXT, DOCX)** are used for chatbots, sentiment analysis.

The format depends on the AI problem.



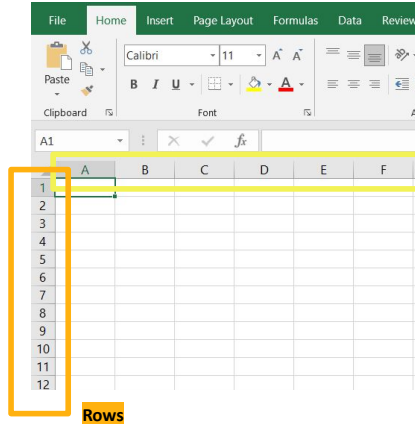
Organizing Collected Data

- After collecting data, we organize it in tables.
- Most datasets are structured as columns and rows.
- Proper organization helps AI detect patterns.



Rows, Columns & Samples (In Data Collection)

- Each Row = One Sample
(one example collected).
- Each Column = One Feature
(attribute collected).
- One special column = Label
(output to predict).
- Each row represents = One sample
(data entry collected during acquisition).



A screenshot of the Google Sheets interface. The 'AIClub' spreadsheet is open. The menu bar shows 'File', 'Edit', 'View', 'Insert', 'Format', 'Data', 'Tools', and 'Add-ons'. The toolbar shows 'Undo', 'Redo', 'Print', 'Zoom', 'Currency', 'Percentage', 'Text color', 'Background color', and 'Default (A)'. The formula bar shows 'fx'. The table has columns A through D and rows 1 through 5. A yellow rectangle highlights the first row (row 1), and an orange rectangle highlights the first column (column A). The table data is as follows:

	A	B	C	D
1	Number of countries visited	Number of years in school	Height (feet)	Who am I?
2	20	15	5.2	Adult
3	2	3	3.5	Child
4	10	12	4.9	Adult
5				

Sample

Each row is an Example for the AI to Train with

Features and Labels

- During Data Acquisition, we decide:
 - Features (Input Data) - What information to collect
 - Example: Study hours, attendance, marks.
 - Label (Output Data) - What we want to predict
 - Example: Pass/Fail, Adult/Child.

	A	B	C	D
1	Number of countries visited	Number of years in school	Height (feet)	Who am I?
2	20	15	5.2	Adult
3	2	3	3.5	Child
4	10	12	4.9	Adult
5				

Sample

Each row is an Example for the AI to Train with

Choosing correct features is very important.

Example Dataset (Collected Data)

- Problem: Predict whether a person is Adult or Child.
- Countries Visited, Years in School, Height are Features collected.
- Label is the last column.

Samples
Each row is an
Example for
the AI to Train
with

<u>Features</u> Features can come in many types (numbers, categories, text etc.) Common real life examples take 100s to 1000s of features!			<u>Labels</u> Each dataset has only one Label column
Number of Countries Visited	Number of Years in School	Height (Feet)	Who am I?
20	15	5.2	Adult
2	3	3.5	Child
10	12	4.9	Adult

This is structured data collected during
Data Acquisition.

Types of Data (While Collecting)

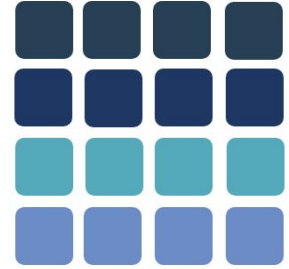
- Structured Data - Organized in tables, easy to analyze.
 - Examples: Marks sheet, survey results.
- Unstructured Data - Not organized in table format.
 - Examples: Images, videos, audio.

UNSTRUCTURED DATA



VS

STRUCTURED DATA

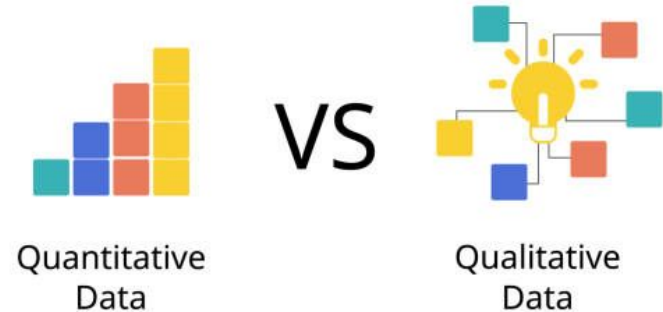


During Data Acquisition, we decide which type is needed.

Quantitative & Qualitative Data (In Collection)

- Quantitative Data is numerical.
 - Example: Marks, age.
- Qualitative Data is descriptive.
 - Example: Feedback, opinions.

Both types may be collected depending on the problem.



Key Takeaway (Data Acquisition Focus)

- ✓ Data Acquisition is about collecting the right data.
- ✓ Data must be stored in proper formats.
- ✓ Data should be organized into rows and columns.
- ✓ Features and labels must be clearly identified.
- ✓ Correct data collection leads to better AI models.

A top-down view of a desk with a light blue background. In the top left, a silver laptop is partially visible. Next to it is a white mouse. Below the mouse, a pair of tortoiseshell glasses lies on the surface. In the bottom left corner, there is a dark brown leather notebook with a white pen tucked into its side.

THANK YOU



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<https://aiclub.world>